

Chapter 2

Lines and Parabolas

The two simplest non-constant functions are the line and the parabola, and each carries a geometric meaning that the algebra alone hides. A line records a fixed rate of change: every step to the right raises the output by the same amount, which is the statement that the right triangles riding along the line are all similar. A parabola records a focusing property: rays arriving parallel to its axis converge to a single point.

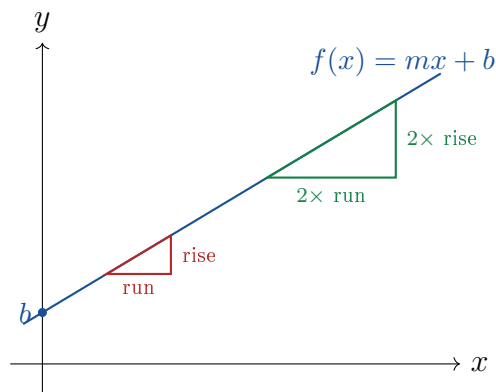
2.1 Lines as Families of Similar Triangles

Take any two points on a non-vertical line and drop the right triangle between them: run horizontally, then rise vertically to meet the line. Choose a different pair of points and build the same triangle. The two triangles have the same angles, because the line meets every horizontal at one fixed angle, so they are similar. Similar triangles have proportional sides, which means the ratio of rise to run is identical no matter which pair of points generated it. That common ratio is the slope.

Definition: Slope. For two points (x_1, y_1) and (x_2, y_2) with $x_1 \neq x_2$, the *slope* of the line through them is

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\text{rise}}{\text{run}}.$$

The picture. A line is a family of similar right triangles stacked along it. Each triangle has a horizontal leg (the run) and a vertical leg (the rise), and the slope m is the fixed ratio rise/run shared by all of them. Doubling the run doubles the rise; the shape of the triangle never changes, only its size.



The number b enters as the height of the line where it crosses the y -axis, the output at input 0. The slope sets the tilt; the intercept sets the height. Together they fix the line completely, which is the content of the slope-intercept form.

Key idea. The slope-intercept form of a line is

$$f(x) = mx + b,$$

where m is the common rise-over-run ratio of every similar triangle along the line and $b = f(0)$ is the height at which the line crosses the y -axis.

Example 2.1.1. (*reading slope and intercept*) The line $f(x) = \frac{3}{4}x - 2$ has slope $\frac{3}{4}$: every run of 4 produces a rise of 3. It crosses the y -axis at $(0, -2)$. From the intercept, stepping right 4 and up 3 reaches $(4, 1)$, a second point on the line.

Common pitfall. Slope is rise over run, not run over rise. The line $f(x) = \frac{3}{4}x - 2$ climbs 3 for every 4 across, a gentle rise. Inverting the ratio to $\frac{4}{3}$ describes a steeper line and a different function.

2.2 Two Points Determine a Line

A point cloud of exactly two points (with different x -coordinates) determines one line and only one. The construction runs in three stages: compute the slope from the two points, write the point-slope form, then rearrange to slope-intercept.

The slope comes straight from the definition. With slope and one known point (x_1, y_1) in hand, every other point (x, y) on the line must produce that same slope against (x_1, y_1) :

$$\frac{y - y_1}{x - x_1} = m.$$

Clearing the denominator gives the point-slope form.

Key idea. The point-slope form of the line through (x_1, y_1) with slope m is

$$y - y_1 = m(x - x_1).$$

Solving for y converts it to slope-intercept form $y = mx + b$.

Example 2.2.1. (*from two points to $f(x) = mx + b$*) Find the line through $(2, 1)$ and $(6, 4)$.

Slope. $m = \frac{4 - 1}{6 - 2} = \frac{3}{4}.$

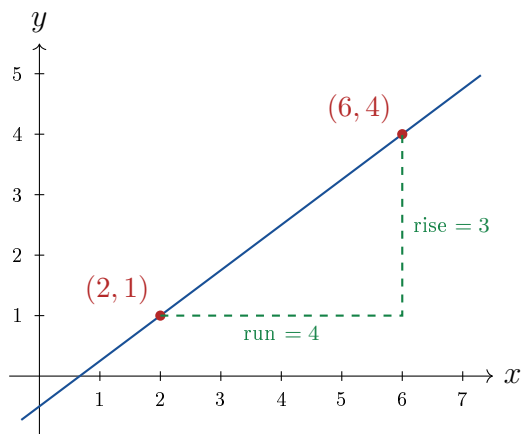
Point-slope. Using $(2, 1)$,

$$y - 1 = \frac{3}{4}(x - 2).$$

Slope-intercept. Expand and solve for y :

$$y = \frac{3}{4}x - \frac{3}{2} + 1 = \frac{3}{4}x - \frac{1}{2}, \quad \text{so} \quad f(x) = \frac{3}{4}x - \frac{1}{2}.$$

Both original points check: $f(2) = \frac{3}{4} \cdot 2 - \frac{1}{2} = \frac{3}{2} - \frac{1}{2} = 1$ and $f(6) = \frac{3}{4} \cdot 6 - \frac{1}{2} = \frac{9}{2} - \frac{1}{2} = 4$.



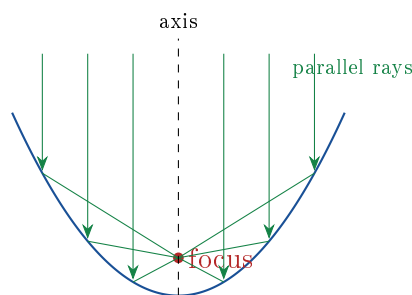
Common pitfall. The slope must use the coordinates in consistent order: the y -difference and the x -difference both run from the same starting point. Subtracting $y_2 - y_1$ over $x_1 - x_2$ flips the sign and tilts the line the wrong way.

Example 2.2.2. (*a horizontal and a vertical case*) Through $(1, 3)$ and $(5, 3)$ the slope is $\frac{3-3}{5-1} = 0$, giving $f(x) = 3$, a horizontal line. Through $(2, 1)$ and $(2, 7)$ the run is $5 - 5 = 0$; the slope formula divides by zero. The relation $x = 2$ is a vertical line, which fails the vertical line test and so is not a function.

2.3 The Parabola as a Focusing Curve

A parabola is not merely a bent line. It is the curve defined by a focusing property: every ray traveling parallel to its axis reflects off the curve and passes through a single point, the focus. This is why headlight reflectors, satellite dishes, and solar collectors are built as parabolas; the geometry concentrates parallel incoming energy at one point, or, run in reverse, sends energy from the focus out as a parallel beam.

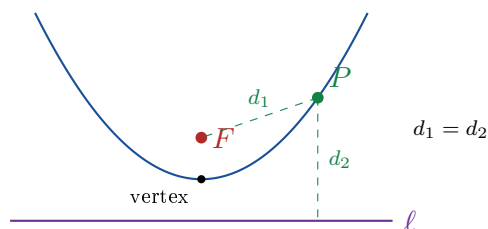
The picture. Parallel rays strike the inside of a parabola and all bounce to the focus. No other curve collects parallel rays to a single point. The algebraic parabola $y = ax^2$ and the optical reflector are the same object viewed two ways.



The focusing property has an equivalent point-and-line description, which is the formal definition of the parabola.

Definition: Parabola. Fix a point F (the *focus*) and a line ℓ not through F (the *directrix*). The *parabola* is the set of all points equidistant from F and ℓ .

So a point and a line determine a parabola through the equidistance condition, just as two points determine a line. The vertex sits halfway between focus and directrix, and the axis is the perpendicular from the focus to the directrix.



2.4 Three Points Determine a Parabola

A vertical-axis parabola is the graph of a quadratic function

$$f(x) = ax^2 + bx + c, \quad a \neq 0,$$

which carries three parameters a , b , c . Three points with distinct x -coordinates impose three conditions, and solving the resulting system fixes the three parameters uniquely. So three points determine the quadratic, in the same sense that two points determine the line.

Key idea. Two points (distinct x) determine a unique line, $f(x) = mx + b$, with two parameters. Three points (distinct x) determine a unique vertical-axis parabola, $f(x) = ax^2 + bx + c$, with three parameters. The count of points matches the count of parameters.

Example 2.4.1. (*fitting a parabola to three points*) Find the quadratic through $(0, 1)$, $(1, 0)$, and $(2, 3)$.

Substitute each point into $f(x) = ax^2 + bx + c$:

$$\begin{aligned}(0, 1) : \quad & c = 1, \\(1, 0) : \quad & a + b + c = 0, \\(2, 3) : \quad & 4a + 2b + c = 3.\end{aligned}$$

From the first, $c = 1$. The second gives $a + b = -1$. The third gives $4a + 2b = 2$, i.e. $2a + b = 1$. Subtracting, $a = 2$, then $b = -3$. The quadratic is

$$f(x) = 2x^2 - 3x + 1.$$

Check $(2, 3)$: $2 \cdot 4 - 6 + 1 = 3$.

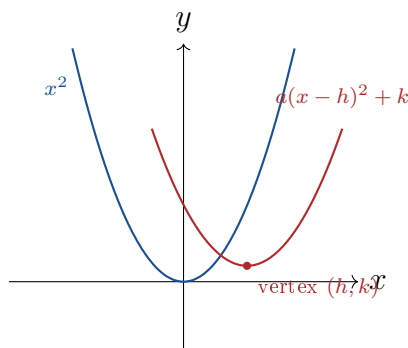
Common pitfall. The three points must have distinct x -coordinates. Two points sharing an x -value would force two outputs at one input, violating the function condition, and the system would have no solution.

2.5 Three Forms of a Quadratic

A quadratic function can be written three ways, each exposing different information. The forms describe the same parabola; converting among them is routine algebra.

Form	Expression	What it shows at a glance
Standard	$f(x) = ax^2 + bx + c$	y -intercept c ; leading behavior a
Factored	$f(x) = a(x - r_1)(x - r_2)$	the roots (x -intercepts) r_1, r_2
Vertex	$f(x) = a(x - h)^2 + k$	the vertex (h, k)

In all three the same a controls the opening: upward for $a > 0$, downward for $a < 0$, and steeper as $|a|$ grows. The vertex form is the transformation form from Week 1 with parent $f(x) = x^2$: h shifts horizontally, k shifts vertically, a scales and possibly reflects.



2.5.1 Standard to vertex: completing the square

Completing the square rewrites $ax^2 + bx + c$ as $a(x - h)^2 + k$ by reconstructing the perfect square hidden in the first two terms. Factor a from the x -terms, add and subtract the square that completes the bracket, then collect.

Example 2.5.1. (*completing the square*) Convert $f(x) = 2x^2 - 12x + 19$ to vertex form.
Factor 2 from the variable terms:

$$f(x) = 2(x^2 - 6x) + 19.$$

The square completing $x^2 - 6x$ uses half of -6 , squared, which is 9:

$$f(x) = 2(x^2 - 6x + 9 - 9) + 19 = 2((x - 3)^2 - 9) + 19.$$

Distribute and collect:

$$f(x) = 2(x - 3)^2 - 18 + 19 = 2(x - 3)^2 + 1.$$

The vertex is $(3, 1)$ and the parabola opens upward, stretched by 2.

The vertex coordinates can also be read off a formula that completing the square produces in general:

$$h = -\frac{b}{2a}, \quad k = f(h).$$

Example 2.5.2. (*vertex by formula*) For $f(x) = 2x^2 - 12x + 19$, $h = -\frac{-12}{2 \cdot 2} = 3$ and $k = f(3) = 18 - 36 + 19 = 1$. The vertex is $(3, 1)$, agreeing with the completed-square result.

2.5.2 Standard to factored: the roots

The factored form exposes the x -intercepts, the inputs where $f(x) = 0$. They come from factoring directly when possible, or from the quadratic formula in general:

$$r_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

The quantity under the root, $b^2 - 4ac$, is the *discriminant*. It records how many real roots exist: two when positive, one (a doubled root at the vertex) when zero, none when negative.

Example 2.5.3. (*standard to factored*) Factor $f(x) = x^2 - x - 6$. Two numbers multiplying to -6 and adding to -1 are -3 and 2 , so

$$f(x) = (x - 3)(x + 2),$$

with roots $r_1 = 3$ and $r_2 = -2$. The discriminant is $(-1)^2 - 4(1)(-6) = 25 > 0$, confirming two real roots.

Example 2.5.4. (*roots by the quadratic formula*) For $f(x) = 2x^2 - 3x + 1$, the discriminant is $9 - 8 = 1$, so

$$r = \frac{3 \pm 1}{4} = 1 \text{ or } \frac{1}{2}.$$

The factored form is $f(x) = 2(x - 1)(x - \frac{1}{2})$. Expanding recovers $2x^2 - 3x + 1$.

2.5.3 Relations among the parameters

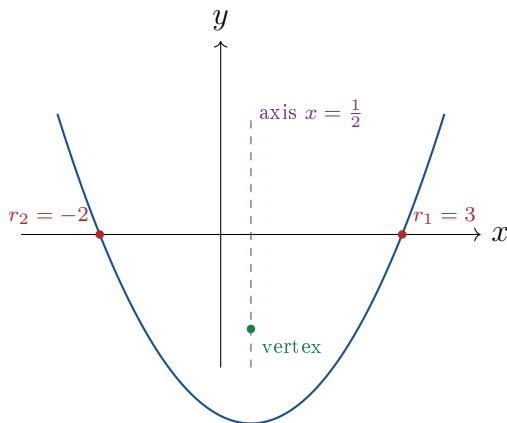
The three forms share structure, and a few relations connect their parameters directly.

Key idea. For $f(x) = ax^2 + bx + c = a(x - r_1)(x - r_2) = a(x - h)^2 + k$:

- the vertex sits at $h = -\frac{b}{2a}$, midway between the roots: $h = \frac{r_1 + r_2}{2}$;
- the product of the roots is $r_1 r_2 = \frac{c}{a}$ and their sum is $r_1 + r_2 = -\frac{b}{a}$;
- the y -intercept is $c = f(0) = a r_1 r_2 = ah^2 + k$.

Example 2.5.5. (*reading the vertex from the roots*) The parabola $f(x) = (x - 3)(x + 2)$ has roots 3 and -2 . Their midpoint is $h = \frac{3+(-2)}{2} = \frac{1}{2}$, so the axis of symmetry is $x = \frac{1}{2}$ and the vertex lies there. Then $k = f(\frac{1}{2}) = (\frac{1}{2} - 3)(\frac{1}{2} + 2) = (-\frac{5}{2})(\frac{5}{2}) = -\frac{25}{4}$. The vertex is $(\frac{1}{2}, -\frac{25}{4})$.

The picture. A parabola is symmetric about the vertical line through its vertex. The two roots sit at equal distances on either side of that axis, so their midpoint is the vertex's x -coordinate. Reading the vertex from the roots is just averaging.



2.6 Exercises

Exercises 2.6

- 2.6.1. Explain why the slope of a line is the same regardless of which two points are used to compute it, referring to similar triangles in your answer.
- 2.6.2. Find the equation of the line through $(-1, 2)$ and $(3, -6)$. Carry it through slope, point-slope, and slope-intercept form, and verify with both points.
- 2.6.3. A line passes through $(4, 1)$ with slope $-\frac{2}{3}$. Write its point-slope and slope-intercept forms, and find its x - and y -intercepts.
- 2.6.4. Decide whether each relation is a function and, if so, give its slope: (a) the line through $(2, 5)$ and $(2, -1)$; (b) the line through $(0, 4)$ and $(7, 4)$; (c) the line through $(1, 1)$ and $(3, 5)$.
- 2.6.5. State the focus-directrix definition of a parabola and explain, in terms of equidistance, why the vertex lies halfway between focus and directrix.
- 2.6.6. Describe the focusing property of a parabola and name two devices that exploit it. Explain which direction the rays travel in each device.
- 2.6.7. Find the quadratic $f(x) = ax^2 + bx + c$ through $(-1, 6)$, $(0, 1)$, and $(2, 3)$ by setting up and solving the linear system.

- 2.6.8. Convert $f(x) = 3x^2 + 12x + 7$ to vertex form by completing the square, and confirm the vertex with the formula $h = -\frac{b}{2a}$.
- 2.6.9. For $f(x) = x^2 - 2x - 15$, find the roots by factoring, the vertex by averaging the roots, and the y -intercept. Sketch the parabola with all three labeled.
- 2.6.10. Use the discriminant to determine the number of real roots of each: (a) $x^2 - 4x + 4$; (b) $2x^2 + x + 3$; (c) $x^2 - 5x + 2$. For any with real roots, give them.
- 2.6.11. A quadratic has roots -4 and 2 and passes through $(0, -16)$. Determine a , then write the function in all three forms.
- 2.6.12. Show, using $h = -\frac{b}{2a}$ and the root-sum relation $r_1 + r_2 = -\frac{b}{a}$, that the vertex's x -coordinate equals the average of the two roots. Explain the connection to the axis of symmetry.